7 A cube of sides 10 cm has a density of 7.8 g cm^{-3}. It floats vertically with one-eighth of its side exposed above the liquid surface.

What is the density of the liquid?

A 6.8 g cm^{-3} **B** 8.9 g cm^{-3} **C** 11.6 g cm^{-3} **D** 62.4 g cm^{-3}

Ans: B

As object is floating, Weight of object = Upthrust

$$\rho_b V_b g = \rho_l V_{submerged} g$$

$$\rho_l = \frac{\rho_b V_b}{V_{submerged}} = \frac{(7.8)(l^3)}{\left(l^2 \times \frac{7}{8} l\right)} = 8.9 \text{ g cm}^{-3}$$

8 The tension in a sample of wire varies with extension as shown in the diagram below.